user["firstName"] ⇝ accessing Object Elements.

↳ remember to use these qoutation while accessing something.

functions ⇩

```
function sum(a, b){
    const sumValue = a+b;
    return sumValue;
}

const value = sum(1, 2);
console.log(value),
```

JavaScript Callbacks → Function as an Argument

```
function sum(num1, num2, fnToCall){
    let result = num1+ num2;
    fnToCall(result);
}
function displayResult(data){
    console.log("Result of the sum is: " + data);
}

function displayResultPassive(data){
    console.log("Sum's result is: " + data);
}
```

```
// you are only allowed to call one function after this
// how will you displayResult of a sum.
const ans = sum(1, 2, displayResult);
```

Another Type.

```
function calculateArithmatic(a, b, arithmaticFinalFunction) {
    const ans = arithmaticFinalFunction(a, b);
    console.log(ans);
}

function sum(a, b) {
    return a + b;
}

const value = calculateArithmatic(1, 2, sum);
console.log(value);
```

↳ will be discussed later in detail.

different/Verbose ways of creating function in javascript.

```
1) function greet(name) {
       return "Hello, " + name;
   }
```

↳ hoisted

↳ is a JS default behavior of moving declaration to the top of their scope (before code execution). Remember declaration are hoisted not intialization.

2) Function Expression (Anonymous or Named)

```
const greet = function(name) {
    return "Hi, " + name;
};
```

↳ not hoisted & since const is there so ; is perferred.

3) Arrow Function (ES6+)

```
const greet = (name) => {
    return "Hey, " + name;
};

// or if single return
const greet = name => "Yo, " + name;
```

4. Immediately Involved Function Expression (IIFE)

```
(function(name) {
    console.log("Welcome, " + name);
})("Eve"); → // Welcome, Eve
```

Common JavaScript Web APIs

```
setTimeout(() => {
    console.log("Runs after 2 seconds");
}, 2000);

const intervalId = setInterval(() => {
    console.log("Runs every 3 seconds");
}, 3000);
```

```
// To stop the interval after 10 seconds.
setTimeout(() => clearTimeout(intervalId), 10000);
// Networking API
fetch("https://jsonplaceholder.typicode.com/posts/1")
    .then(response => response.json())
    .then(data => console.log("fetched data:", data));
```

JavaScript Callback ⇝